30.4 A 28nm 106.85TOPS/W and 77.68TFLOPS/W CIM Macro with Stage-Wise-Enabled Lossless Compressors Based on Sign-Bit-Embedded Transition-Counting-Lines for Edge-AI Devices

Lichen Feng, Yunlong Liu, Lin Wu, Dong Li, Zhangming Zhu

Xidian University, Xi'an, China

Abstract

This paper proposes a bit-parallel digital CIM macro featuring a lossless compressor based on transition-counting-lines (TCLs) for bit-column addition. The bus TCL incorporates sign-bit extension to support signed 2's-complement MAC operations with minimal area overhead. A stage-wise enabling circuit prevents unnecessary toggling of adders, improving energy efficiency by 42%. Leveraging these techniques, the proposed CIM macro, which is fabricated in 28nm CMOS, achieves 106.85TOPS/W for INT8 operations and 77.68TFLOPS/W for BF16 operations.

SRAM CIM macros improve energy efficiency for edge-AI devices. In contrast to "bit-serial" CIM macros [1], "bit-parallel" CIM directly matches the data format of computing systems, eliminating the parallel-to-serial conversion and specialized weight mapping strategies [2,3]. It can also reduce the MSB toggle rate [4], thereby lowering power consumption. Current bit-parallel CIM macros have achieved notable progress [5-7], however, as illustrated in Fig. 30.4.1, several challenges remain: (1) For 8b MAC, "bit-parallel" CIM suffers from routing congestion [7]. Some approaches concatenate multi-cycle computation results from low-precision MAC units (e.g., INT4) [6][8], while others employ ADCs as local compressors [5][9] to perform bit-column addition. A tradeoff between throughput and area overhead is required. Approximate compressors [6,10] can reduce resource consumption, but with accuracy loss. (2) Handling sign bits for 2's-complementary MAC in these macros often demands large extra area [6][8] or additional cycles [11]. (3) During multiplication and local compression before generating the final outer-product results, intermediates induce unnecessary toggling in high-fan-in adders, resulting in power inefficiency [12].

To address these challenges, this paper presents a fully-digital bit-parallel CIM macro with three key features: (1) Lossless digital local compressors based on transition-counting-lines (TCLs), realizing sparsity-adaptive and area-efficient (minimized transistor count) bit-column addition for outer-product matrix; (2) A bus TCL embedding sign-bit extension for signed 2's-complementary MAC, incurring only 2.95% extra area without additional delay; (3) A shared local 10-phase generator, enabling TCLs and adder trees to operate in a stage-wise manner within a computation cycle, thereby avoiding unnecessary toggling of adders and improving energy efficiency by 42%. A 28nm 32kb CIM macro based on TCLs is fabricated to support INT8 and BF16 MAC operations, achieving energy efficiencies of 106.85TOPS/W for INT8 and 77.68TFLOPS/W for BF16.

Figure 30.4.2 illustrates the architecture of the presented TCL-based CIM macro, which includes a global controller (Ctrl.), an SRAM read/write controller, four CIM banks, a global input buffering unit, and an exponent post-processing unit for BF16. Each CIM bank consists of a local 10-phase generator, an Out-Block Adder Tree, and four CIM blocks (each containing 16 CIM cells and an in-block adder tree). The in-block adder tree adds the 16 16b products from the 16 CIM cells, and the out-block adder tree adds the four 20b sums from the four In-Block Adder Trees, generating the 22b MAC Value (MACV). A CIM cell integrates a 16×8b 6T-SRAM for storing unsliced weights and a computing cell that performs 8b×8b bit-parallel multiplication. The computing cell comprises 64 multiply-multiplex units (MMUs), 15 counters (CNTs), and a combine adder, reading an 8b weight per cycle from the SRAM. The MMUs at the same bit-position are connected hierarchically, with a CNT at the end to form a TCL, performing bit-column additions. Following the outer-product rule, 15 TCLs are used for 8b×8b multiplication, and the 15 bit-column sums are accumulated by the Combine Adder to obtain a 16b product. The MMUs for sign-bits in the bus TCL (TCL-14) are reused across TCL-7~14 as embedded sign-bit extension for 2's-complementary multiplication. A clock cycle includes 10 phases: phases 0~7 (Ph0~7) sequentially enable MMUs in each TCL, Ph8 enables the Combine Adder, and Ph9 enables the In-Block Adder Tree. The CIM macro supports both INT8 and BF16 MAC operations. In INT8 mode, 64 features (XIN, 8b) are applied to four CIM banks, which hold 64×4 pre-read 8b weights, yielding four 22b MACVs per clock cycle. In BF16 mode, the exponents of the input features are pre-aligned [1], and the CIM banks store mantissas of the pre-aligned weights. The four mantissa MACVs obtained in the four CIM banks are post-processed to generate the FP32 output. These three stages are pipelined to generate the FP32 output per clock cycle.

Figure 30.4.3 illustrates the circuit structures and operations of the 10-phase generator and the TCL. The 10-phase generator comprises ten identical Rising-edge Delay Cells (RDCs) connected in cascade. V_{rst} controls each RDC's output state, while V_{delay} regulates its input-output delay. V_{rst} is synchronously asserted high (logic 1) with the rising edge of the main clock (Clk) and then quickly deasserted low (logic 0). When V_{rst}=1, all RDC outputs (Ph0~9) are set to 0. Ph0 is driven high when V_{rst}=0. After the delay of RDC #0 (Δt_{r0}), Ph1 is driven high as well. This process propagates through the 10 RDCs, driving Ph0~9 to high sequentially, generating the 10 phases as shown in Fig. 30.4.3. The TCL is composed of MMUs and a CNT. Each MMU comprises a Multiply section and a Multiplex section. The Multiplex sections of the MMUs in a TCL are cascaded: the output (MOut) of each MMU serves as the input (MIn) of the next MMU. The MIn of the first MMU is tied to V_{DD} and the MOut of the last MMU is connected to the input of the CNT. The Multiply section determines whether *Path1* or *Path2* is activated in the Multiplex section, depending on the values of Phk, XIN[m], and W[n] (k, m, n=0~7). Only when phase Phk=1 and P_{mn}=XIN[m]×W[n]=1 is *Path1* activated. In that case, MOut is the inversion of MIn, producing a transition (edge) that CNT counts. Taking TCL-9 with eight MMUs as an example, where the bit-products are {1, 1, 0, 0, 1, 0, 0, 1}, respectively: During Δt_{r0} (Ph0=1 and Ph1~9=0), only the first MMU (corresponding to P_{07}) activates *Path1*; since the MIn of the first MMU is tied to V_{DD}, *Path1* yields a falling-edge-transition at MOut. The remaining MMUs activate *Path2* and pass the signal unmodified, allowing the falling edge to propagate to the CNT and increment the count by 1. Similarly, transitions occur during $\Delta t_{r1,4,7}$. In total, the CNT detects four transition edges as the bit-column sum. The eight bit-products are encoded as the 4b count, forming an 8-to-4 lossless compression. Ph8 and Ph9 remain low during Ph0~7, so the Combine Adder and Adder Tree remain inactive, reducing dynamic power consumption.

Figure 30.4.4 illustrates how the bus TCL implements embedded sign-bit extension, and the enabling-circuit structures for Combine Adder and In-Block Adder Tree. For 2's-complementary signed multiplication, TCL-14 forms the bus TCL by connecting P_{m7}, m=0~7. P_{07} is used by each of TCL-7~14 (thus reused 8×), P_{17} is used by each of TCL-8~14 (reused 7×), and so forth, thereby realizing sign-bit extension. To implement the complete 2's-complementary multiplication rule, the inverted W_n's (n=0~7) are applied to the invert-bit MMUs, respectively, and the CNT in TCL-7 is initialized to XIN[7]. For unsigned computation, we only need to disconnect P_{m7} (m=0~7) in TCL-14, multiplex W_n's to the invert-bit MMUs, and initialize the CNT in TCL-7 to 0. Switching between the two modes requires only seven transmission gates (TGs) and eight 2-to-1 multiplexers, with an additional area overhead of 2.95%. Before Ph8, the inputs of the combine adder and the in-block adder tree are forced to ground using single-pole double-throw switches, preventing the propagation of spurious toggling. As shown in Fig. 30.4.2, the In-Block Adder Tree is of four stages. Each stage can include phase-controlled switches to improve energy efficiency, while they increase area overhead. We simulated the area and efficiency, versus the number of enabling stages. As a tradeoff, one enabling stage is selected for Combine Adder and one for In-Block Adder Tree, achieving a 42% energy efficiency improvement with only 6% area efficiency degradation.

Figure 30.4.5 illustrates the performance of the presented design. The 8b bit-parallel MAC computing cells are compared in terms of MAC cell FoM (Energy Efficiency/Transistor Count). In comparison with standard digital bit-multiplication with standard digital compressors, analog bit-multiplication with two 3b ADCs as compressors [5], and the design [6] employing two 4b bit-parallel MAC cells with ripple carry adders, the presented MAC cell with TCLs achieves the FoM improvement by 1.66 to 1.87×. To better demonstrate the impacts of sign-bit extension and stage-enabling, two additional implementations are evaluated: (1) a baseline unsigned MAC cell without bus TCLs or the adder-enabling circuit, and (2) a signed MAC cell without the enabling circuit. Relative to the baseline design, connecting P_{m7}'s (m=0~7) for bus TCL and increasing CNT bit-widths in signed MAC cell consume 1.03× area and more power. With the introduction of the two-stage adder-enabling circuit, the final fabricated design achieves a 1.35× energy efficiency with only a 1.09× area overhead compared to the baseline unsigned design, and a 1.42× improvement with a 1.06× area overhead compared to the signed design without adder enabling. The 10k Monte Carlo simulation results of the transition delay of MMUs (Δt_{MMU}) and the transition delay of RDCs ($\Delta t_{r0\sim7}$, annotated as Δt_{RDC}) indicate that, Δt_{MMU} is consistently smaller than Δt_{RDC}, ensuring correct TCL operation under process and supply voltage variations. The energy efficiency of the presented signed MAC macro is measured across various input toggle rates and input sparsity levels, with the 50% weight sparsity. A peak performance of 106.85TOPS/W is achieved.

Figure 30.4.6 shows the measurement results of the TCL-based CIM chip, which is fabricated in a 28nm CMOS technology. The measured access time is 4.1ns at 0.9V for BF16 inputs and weights, and FP32 outputs. The measured maximum energy and area efficiency are 77.68TFLOPS/W and 1.24TFLOPS/mm² for BF16 MAC operation under 0.55V

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and 0.9V supply voltages, respectively, with 90% input sparsity and 10% toggle rate, while the weight sparsity is 50%. By adopting the sign-bit-embedded TCL and stage-wise-enabling, the presented bit-parallel CIM macro improves the output precision×TOPS/W×TOPS/mm² (FoM1) by 1.25 to 2.88× and the TFLOPS/W×TFLOPS/mm² (FoM2) by 2.20 to19.9×, in comparison to prior signed CIM macros [2,3,6,7]. In BF16 mode, this work achieves an 81.05% inference accuracy when used on ViT-B @ImageNetV2 and an 81.19% inference accuracy when used on ResNet50 @ImageNetV2. Figure 30.4.7 presents a die micrograph and a summary of chip performance.

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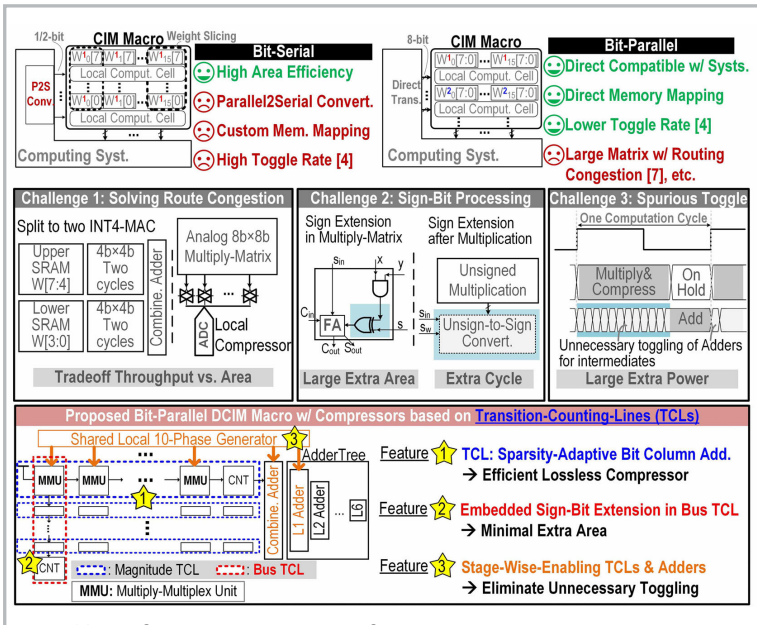


Figure 30.4.1: Challenges of bit-parallel CIM macro and the proposed solution.

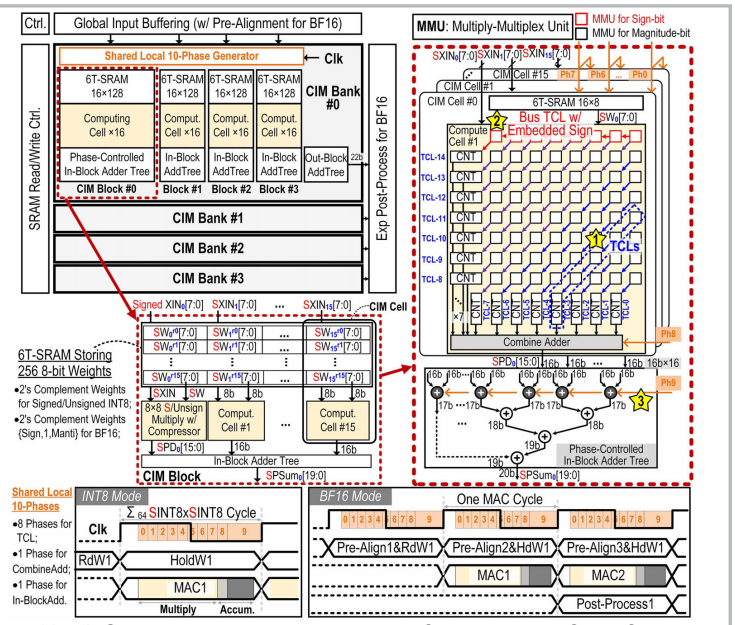


Figure 30.4.2: Overall architecture of the proposed TCL-based digital-SRAM CIM macro.

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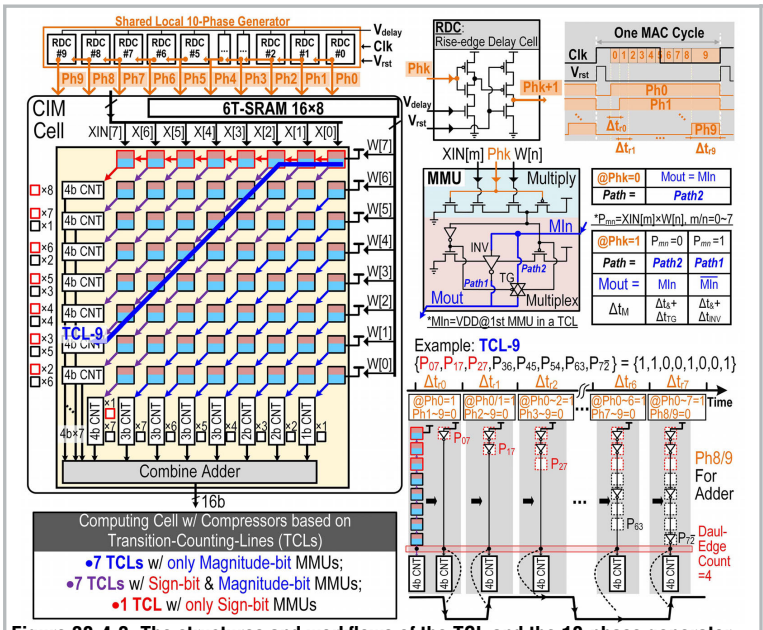


Figure 30.4.3: The structures and workflows of the TCL and the 10-phase generator.

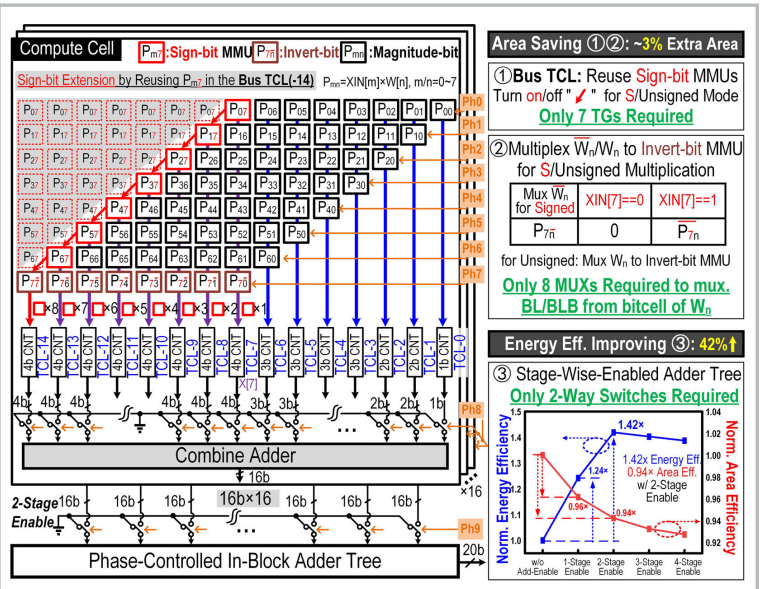


Figure 30.4.4: Embedded sign-bit extension in bus TCL-14 and the enabling-circuit structure for Combine Adder and In-Block Adder Tree.

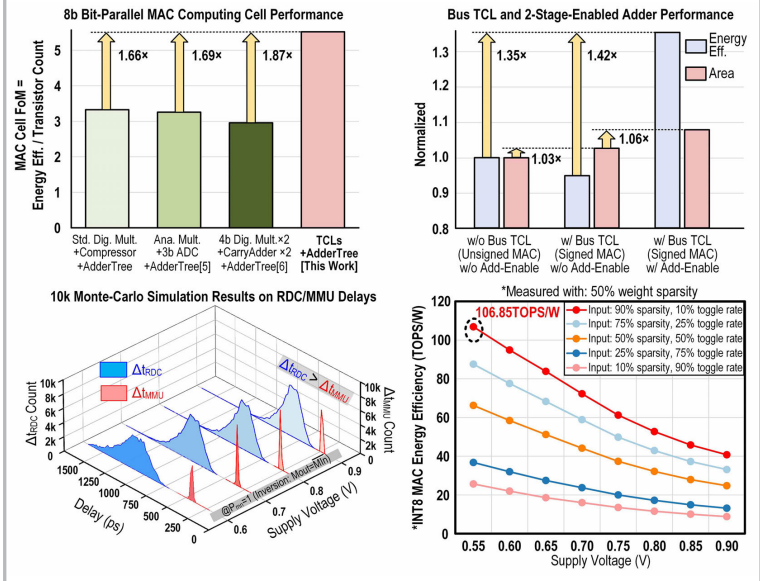
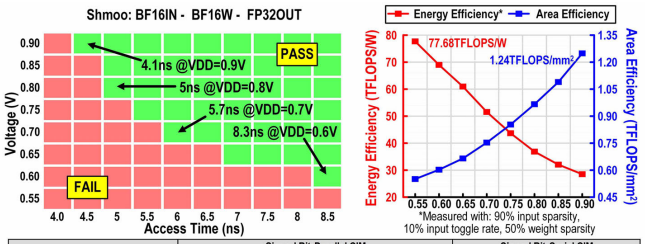


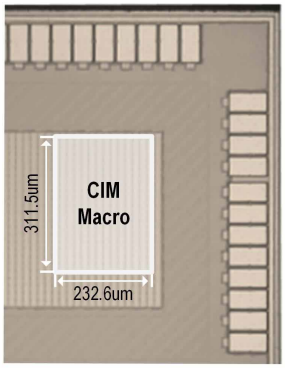
Figure 30.4.5: Simulated MAC computing cell performance and measured energy efficiencies of the proposed CIM macro.



	Signed Bit-Parallel CIM						Signed Bit-Serial CIM			
	This Work		ISSCC 2025[6]		ISSCC 2023[7]		ISSCC 2025[2]		ISSCC 2025[3]	
Technology	28nm		28nm		22nm		28nm		28nm	
CIM Type	Digital		Digital		Hybrid		Digital		Hybrid	
Supply Voltage	0.55-0.9		0.55-0.9		0.6-0.9		0.55-0.9		0.6-0.9	
Macro Size	32Kb		32Kb		832Kb		64Kb		64Kb	
MAC Cell Count	256		64		4096		256		1024	
Area (mm²)	0.072		0.043		18		0.136		0.108	
Memory Density (Kb/mm²)	442.09		744.19		46.22		470.59		580	
Input Precision (bit)	INT8	BF16	INT8	BF16	BF16		INT8	BF16	INT8	
Weight Precision (bit)	INT8	BF16	INT8	BF16	BF16		INT8	BF16	INT8	
Output Precision (bit)	INT22	FP32	INT22	-	FP32		INT20	FP32	INT21-25	
Number of IN Channels (N_{ACCUM})	64		64		128		16		32-512	
Access Time (ns)	4.1		2.1	5.3	6.4		4.8	5.4	12	
INT8 Energy Eff. (TOPS/W)	**40.74 - 106.85** *1		36.5 - 104		-		25.47 - 90.15		21.04-67.8	
BF16 Energy Eff. (TFLOPS/W)	**28.49 - 77.68** *1		26.6 - 75.8		62.14 - 70.21		17.83 - 62.84		-	
INT8 Area Eff. (TOPS/mm²)	**1.73**		1.42 *2		-		0.784		0.37-1.57	
BF16 Area Eff. (TFLOPS/mm²)	**1.24** *3		0.345 *2,3		0.069		0.697		-	
FoM1 *4	**4066.71**		3248.96		-		1413.55		2661.15	
FoM2 *5	**96.32**		26.15		4.84		43.80		-	
Inference Accuracy Loss	0.02% BF16×BF16 ViT@ImageNet		0.1% BF16×BF16, VGG19@CIFAR100		0.43% BF16×BF16 ResNet18@ImageNet		0.016% BF16×BF16 ViT@ImageNet		1.75% INT8×INT8 ViT@ImageNet	

*1 : Measured under 90% input sparsity, 10% toggle rate, 50% weight sparsity; *2: Recalculated: (2 Ops × MAC Cell Count) / Access Time / Area;
*3: Including exponent computation circuits; *4 : FoM1 = Output Precision × TOPS/W × TOPS/mm²; *5: FoM2 = TFLOPS/W × TFLOPS/mm².

Figure 30.4.6: Measurement results and comparison table.



CHIP SUMMARY		
Technology	28nm CMOS	
Cell Structure	Compact 6T+TCLs	
Macro Size	32Kb	
Macro Area (mm²)	0.072 (232.6um×311.5um)	
Input Precision (bit)	INT8	BF16
Weight Precision (bit)	INT8	BF16
Number of IN Channels (N_{ACCU})	64	64
Output Precision (bit)	INT22	FP32
Supply Voltage (V)	0.55-0.9	
Output Ratio	1	
Access Time (ns)	4.1 @ 0.9V	
[1]Average Energy Efficiency (TOPS/W) (TFLOPS/W)	24.76-65.26	17.32-48.17
[2]Peak Energy Efficiency (TOPS/W) (TFLOPS/W)	40.74-106.85	28.49-77.68
Area Efficiency (TOPS/mm²) (TFLOPS/mm²)	1.73	[3] 1.24
Memory Density (Kb/mm²)	442.09	
Inference Accuracy [4]ResNet50 @ImageNet1k_V2	80.10%	80.19%
Inference Accuracy [5]ViT-B @ImageNet1k_V2	79.50%	81.05%

[1] Measured under 50% input sparsity, 50% input toggle rate, 50% weight sparsity
[2] Measured under 90% input sparsity, 10% input toggle rate, 50% weight sparsity
[3] Including exponent computation circuits.
[4] Software baseline (FP32) is 80.34%.
[5] Software baseline (FP32) is 81.07%.

Figure 30.4.7: Die micrograph and summary of key performance metrics.

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